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SECJ2154 OBJECT-ORIENTED PROGRAMMING

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MINI PROJECT REPORT:
HOMESTAY TAMAN RIA BOOKING SYSTEM

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GROUP 5

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1.0 Introduction

The rapid growth of the tourism industry has increased the demand for efficient, user-friendly homestay booking systems. However, many homestay businesses still rely on manual methods such as spreadsheets or messaging apps to manage reservations, leading to inefficiencies and frequent issues such as double bookings, outdated availability, and communication errors. To address these challenges, our project proposes the development of the Homestay Taman Ria Booking System, an object-oriented application designed to streamline the booking process for both homestay owners and customers.

This system aims to centralize homestay management by integrating features such as real-time availability tracking, customer reservation management, and automated booking confirmation. It will allow users to view available properties, check prices, compare options, and make informed booking decisions through a structured interface. The proposed solution will be developed using Java, implementing core object-oriented programming principles like inheritance, polymorphism, abstraction, and interface usage.

By providing a systematic and automated platform, this project not only reduces administrative burdens for owners but also improves the overall user experience for customers. It demonstrates practical applications of OOP concepts while addressing real-world problems in the homestay rental industry.

2.0 Problem Statement

Currently, many homestay businesses operate without a centralized system to manage property listings, availability, and customer bookings. This often leads to challenges such as double bookings, poor communication between customers and owners, and difficulties in tracking reservations.

1. Lack of Centralized Homestay Management

Currently, homestay owners may manage their listings manually using spreadsheets, social media platforms or messaging apps. This fragmented approach can result in:

- Inconsistent availability data (e.g., double bookings or outdated listings).
- Difficulty managing multiple properties, especially for owners with several homestays.
- No integrated way to track all bookings, customer details, or revenue.

2. Manual and Inefficient Booking Process

Without an automated booking system, customers typically need to:

- Manually contact the owner to check availability.
- Wait for confirmation, leading to delayed responses.
- Face miscommunication issues, such as booking errors or mismatched dates. This leads to poor user experience and potential loss of bookings.

3. Absence of Date and Availability Management

Homestays often face issues where:

- Owners forget to update available dates.
- Customers book overlapping dates.
- There's no automated system to add or remove available dates, leading to double bookings or booking denials.

4. Limited Visibility for Customers

Customers lack a platform to search, view, and compare homestays. With no structured information (like name, location, price, available dates), it becomes difficult to:

- Find suitable homestays quickly.
- Make informed decisions based on consistent criteria.
- See real-time availability of homestays.

3.0 Objectives

In the current landscape, many homestay businesses continue to operate without a centralized digital infrastructure to manage property listings, booking availability, and customer reservations. This lack of integration often results in operational inefficiencies such as double bookings, outdated availability information, poor customer-owner communication, and inconsistent record-keeping. To address these persistent challenges and improve homestay management's overall reliability and user experience, this project proposes developing an object-oriented booking system tailored for homestay operators and customers.

In alignment with this goal, the following objectives have been established:

1. To design and implement a centralized homestay booking system

- This objective focuses on developing an integrated software solution that consolidates homestay listings, reservation records, and user interactions into a cohesive platform. The aim is to eliminate the fragmentation caused by manual processes and third-party messaging tools.

2. To automate the reservation and availability management process

- The system shall automate key booking functions, including availability checks, reservation confirmations, and date blocking. This ensures real-time synchronization of booking data, significantly reducing the risk of human error and double bookings.

3. To enhance the user experience for both homestay owners and customers

- A core objective is to provide a structured, intuitive interface that enables customers to browse available properties, compare prices, and make informed booking decisions. Similarly, the system aims to assist owners in managing their listings and reservations with minimal manual intervention.

4. To implement object-oriented design principles in solving real-world problems

- The system will be developed using object-oriented programming (OOP) methodologies, including abstraction, inheritance, polymorphism, and interface-based design. This objective demonstrates how foundational programming paradigms can be applied to create scalable and maintainable software systems.

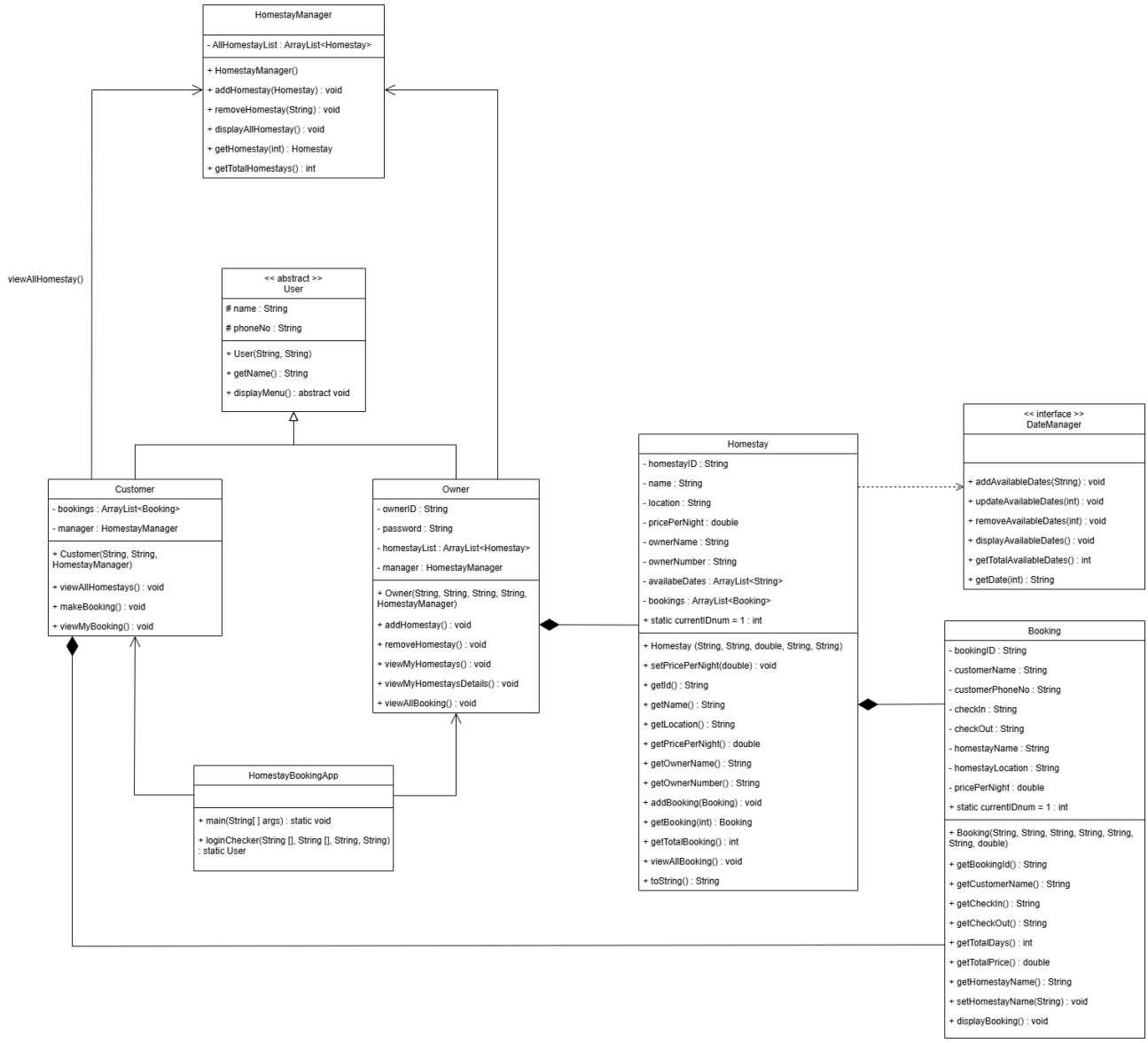
5. To ensure data consistency, integrity, and traceability throughout the booking lifecycle

- The project seeks to provide robust mechanisms for tracking customer interactions, booking history, and homestay availability, with an emphasis on data accuracy, auditability, and structured data storage.

6. To lay the groundwork for future system extensibility and integration

- The design and implementation will consider future enhancements, such as graphical user interfaces (GUI), payment gateway integration, and multi-platform deployment, thereby enabling scalability and long-term system sustainability.

4.0 Project Design (Class Diagram)



Person In Charge	Class
Mior Danial Hakim	User, Customer, Owner
Tan Jian Ming	Homestay, Booking
Muhammad Syahmi	Booking
Muhammad Hafizi Syamir	HomestayManager, DateManager

5.0 Techniques/Software/Tools Being Used

	ENUM	✓	**ASSOCIATION***
✓	ARRAY LIST / VECTOR	✓	COMPOSITION
✓	ARRAY		AGGREGATION
	ARRAY OF OBJECT	✓	**INHERITANCE***
	FILE*	✓	STATIC METHOD
✓	**POLYMORPHISM***	✓	**ABSTRACT CLASS***
✓	**INTERFACE CLASS***	✓	**EXCEPTION HANDLING***
	GUI APPLICATION		Others :

- ***compulsory – also comments, style etc**

Tool used : VS Code, Drawio

6.0 Project Output (Screenshot)

Main Menu :

```
=====================================
      TAMAN RIA HOMESTAY BOOKING
=====================================

1. Owner
2. Customer
3. exit

choose category:
```

Owner :

```
Login Sucessfull!
=====================================
              OWNER MENU
=====================================
1. Add Homestay
2. Remove Homestay
3. View My Homestay
4. View All Booking
Choose option (0 to Main Menu):
```

```
=====================================
        ADDING NEW HOMESTAY
=====================================


Enter Homestay Name (press 0 to cancel): D'Homestay
Enter Location: Tampoi
Enter Price Per Night (RM): 50
Enter Number of Available Dates: 4

**Date Format DD/MM/YYYY**
Date 1: 22/06/2025
Date 2: 23/06/2025
Date 3: 24/06/2025
Date 4: 25/06/2025

Homestay Added!
```

```
=====================================
           REMOVE HOMESTAY
=====================================

1) D'Homestay

Enter no. on the list to remove a homestay (press 0 to cancel):
```

```
=====================================
         LIST OF OWNED HOMESTAY
=====================================

------------ Homestay 1 ------------
Homestay ID    : H001
Name           : D'Homestay
Location       : Tampoi
Price Per Night: RM 50.0

Available Dates:
1. 22/06/2025
2. 23/06/2025
3. 24/06/2025
4. 25/06/2025

Edit Homestay? (Y/N):
```

```
=====================================
           VIEW ALL BOOKINGS
=====================================

=====================================
         HOMESTAY: D'Homestay
=====================================

------------ Booking 1 ------------
Booking ID    : B001
Customer Name : Ahmad
Customer Phone: 0123456789
Homestay Name : D'Homestay
Location      : Tampoi
Check-In Date : 23/06/2025
Check-Out Date: 24/06/2025
Total Days    : 1
Total Price   : RM 50.0 (RM 50.0/night)


Please press Enter to continue...
```

Customer:

```
=====================================
             CUSTOMER MENU
=====================================
1. View All Homestays
2. Make Booking
3. View My Booking
Choose option (0 to Main Menu): 1
```

```
=====================================
          VIEW ALL HOMESTAYS
=====================================

------------ Homestay 1 ------------
Name           : D'Homestay
Location       : Tampoi
Price Per Night: RM 50.0
Owner          : Labu
Contact No     : 0198765432
```

```
=====================================
             MAKE BOOKING
=====================================

------------ Homestay 1 ------------
Name           : D'Homestay
Location       : Tampoi
Price Per Night: RM 50.0
Owner          : Labu
Contact No     : 0198765432

Please press Enter to continue...

Enter no. on the list to select homestay (press 0 to cancel): 1

=====================================
          SELECTED HOMESTAY
=====================================
Name           : D'Homestay
Location       : Tampoi
Price Per Night: RM 50.0

Available Dates:
1. 22/06/2025
2. 23/06/2025
3. 24/06/2025
4. 25/06/2025

Enter no. on the list for Check In date (press 0 to cancel): 2
Enter no. on the list for Check Out date (press 0 to cancel): 3

Your booking has been made!

Please press Enter to continue...
```

```
=======================================
            VIEW MY BOOKINGS
=======================================

------------ Booking 1 ------------
Booking ID     : B001
Customer Name  : Ahmad
Customer Phone: 0123456789
Homestay Name  : D'Homestay
Location       : Tampoi
Check-In Date  : 23/06/2025
Check-Out Date: 24/06/2025
Total Days     : 1
Total Price    : RM 50.0 (RM 50.0/night)

Please press Enter to continue...
```

7.0 Conclusion

The Homestay Taman Ria Booking System successfully demonstrates how object-oriented programming (OOP) principles can be applied to solve real-world problems, particularly in the domain of digital booking and reservation management. Through the implementation of this project, the team has managed to create a functional, modular, and maintainable application that streamlines the process of managing homestay bookings for both owners and customers.

One of the primary goals of this project was to address the inefficiencies observed in traditional homestay management, especially those relying on manual tools like spreadsheets and messaging platforms. These methods often result in miscommunication, double bookings, and outdated availability information. The system developed in this project provides a centralized platform that enables property owners to input and update homestay details and available dates, while allowing customers to browse listings and make bookings with confidence.

Key object-oriented features such as abstraction, inheritance, polymorphism, interfaces, and encapsulation were thoroughly applied throughout the system design. The use of an abstract User class with subclasses Customer and Owner shows proper use of inheritance. The BookingInfo interface ensures that any booking class will provide a standardized confirmBooking() method, demonstrating interface usage and promoting consistency. Additionally, exception handling, array lists, static methods, and composition were all integrated effectively to support system functionality and robustness.

From a structural perspective, the division of responsibility among different classes—such as HomestayManager, DateManager, and Booking—demonstrates a modular approach that facilitates future scalability. For example, future enhancements such as graphical user interfaces (GUI), database connectivity, or integration with payment gateways can be added with minimal changes to the core system.

The system also significantly improves the user experience by providing customers with a straightforward way to view homestay details, check real-time availability, and make reservations. At the same time, it simplifies administrative tasks for homestay owners by centralizing property and booking management in one application. The date management system helps to minimize errors and ensures that users are clearly informed about available and booked dates .

This project also emphasizes the importance of proper software design planning, from early stages like class diagram creation and problem analysis, to implementation and testing. Each team member contributed to specific components, ensuring a collaborative development process. Tools such as VS Code and Draw.io were effectively used to code, debug, and visualize system structure and flow.

In conclusion, the Homestay Taman Ria Booking System meets its intended objectives by providing an efficient, user-friendly, and maintainable solution for managing homestay bookings. It not only addresses current industry shortcomings but also highlights the versatility and practicality of object-oriented programming in software development. The system is a solid foundation for further research, refinement, and potential deployment in a real-world environment. The project serves as a meaningful academic exercise and a potential starting point for future enhancements in the field of hospitality and property management systems.

8.0 Appendices (Java File)

Java File and Presentation Slides : **Mini Project**